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(54) **VOLTAGE LEVEL SHIFTER WITH
PROGRAMMABLE HIGH SUPPLY VOLTAGE
AND HIGH SUPPLY VOLTAGE-DEPENDENT
VARIABLE LOW SUPPLY VOLTAGE**

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19/00384; **H03K 19/017581**;

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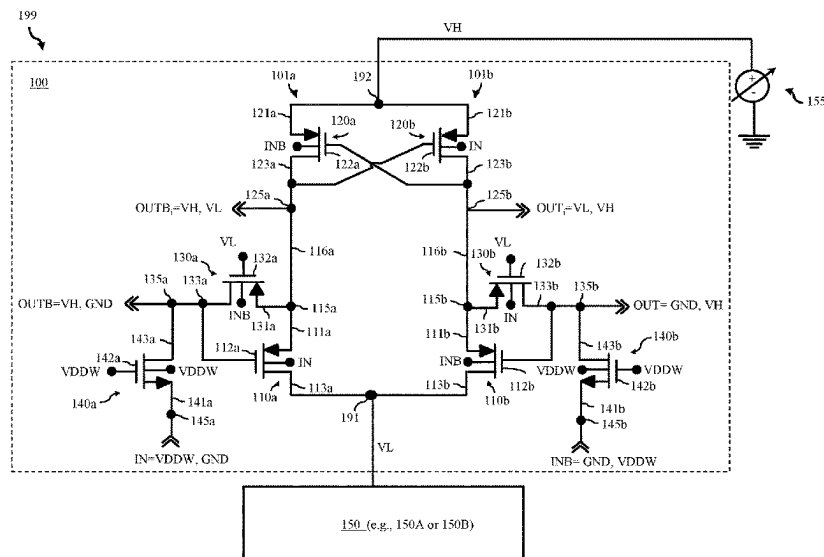
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(57) **ABSTRACT**

A disclosed circuit structure includes a voltage level shifter connected to a variable voltage generator for receiving a low supply voltage and to a programmable voltage generator for receiving a high supply voltage that is higher than the low supply voltage. The high supply voltage is programmable to one of multiple different voltage levels. The variable voltage generator tracks the voltage level of the high supply voltage and outputs the low supply voltage based thereon. The high supply voltage is tracked either directly or using the trim bit signal employed by the programmable voltage generator to program the high supply voltage. The low supply voltage is high when the high supply voltage is high to ensure operation within the safe operating area and is low when the high supply voltage drops below some threshold level to ensure operability and, optionally, improve operating speed.

20 Claims, 5 Drawing Sheets



(58) **Field of Classification Search**

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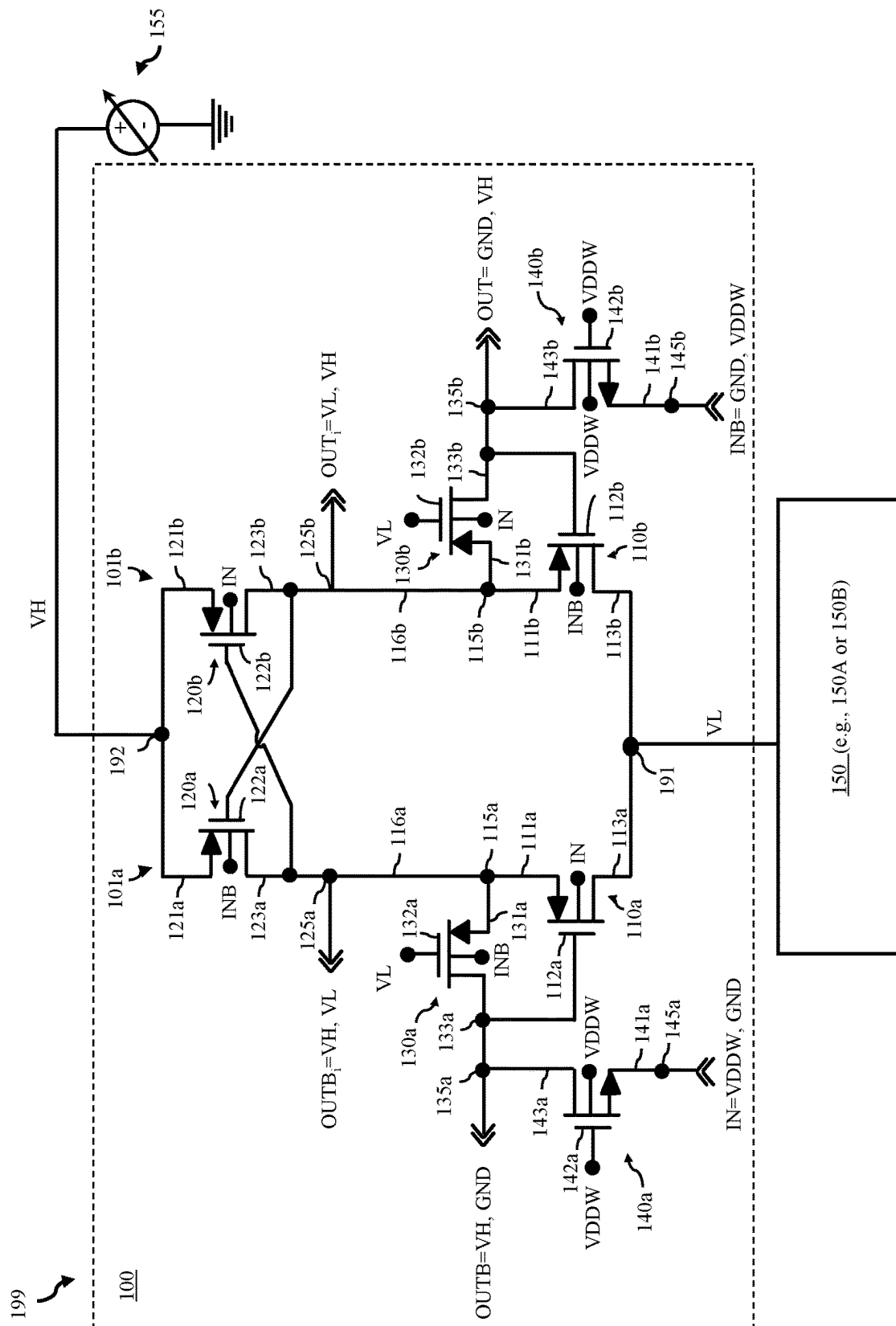


FIG. 1

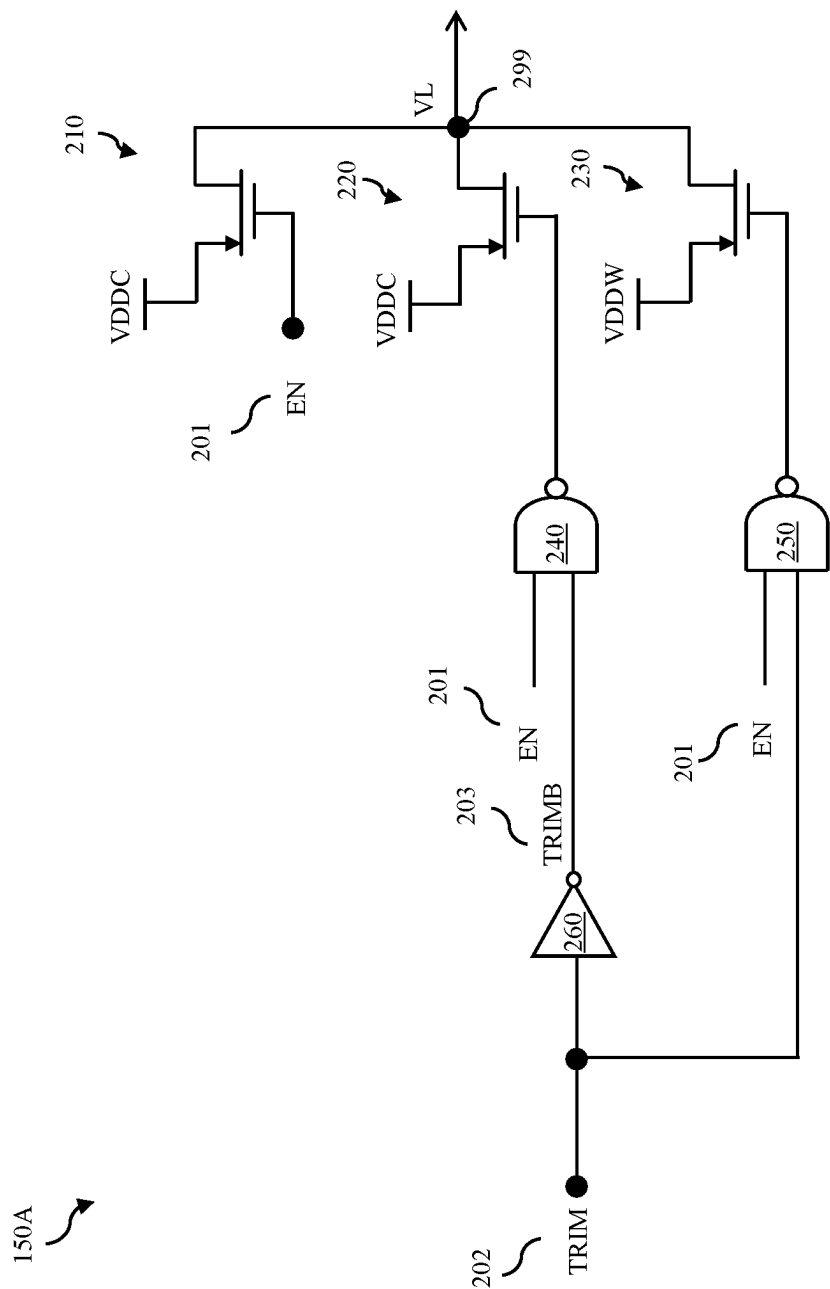


FIG. 2

150B

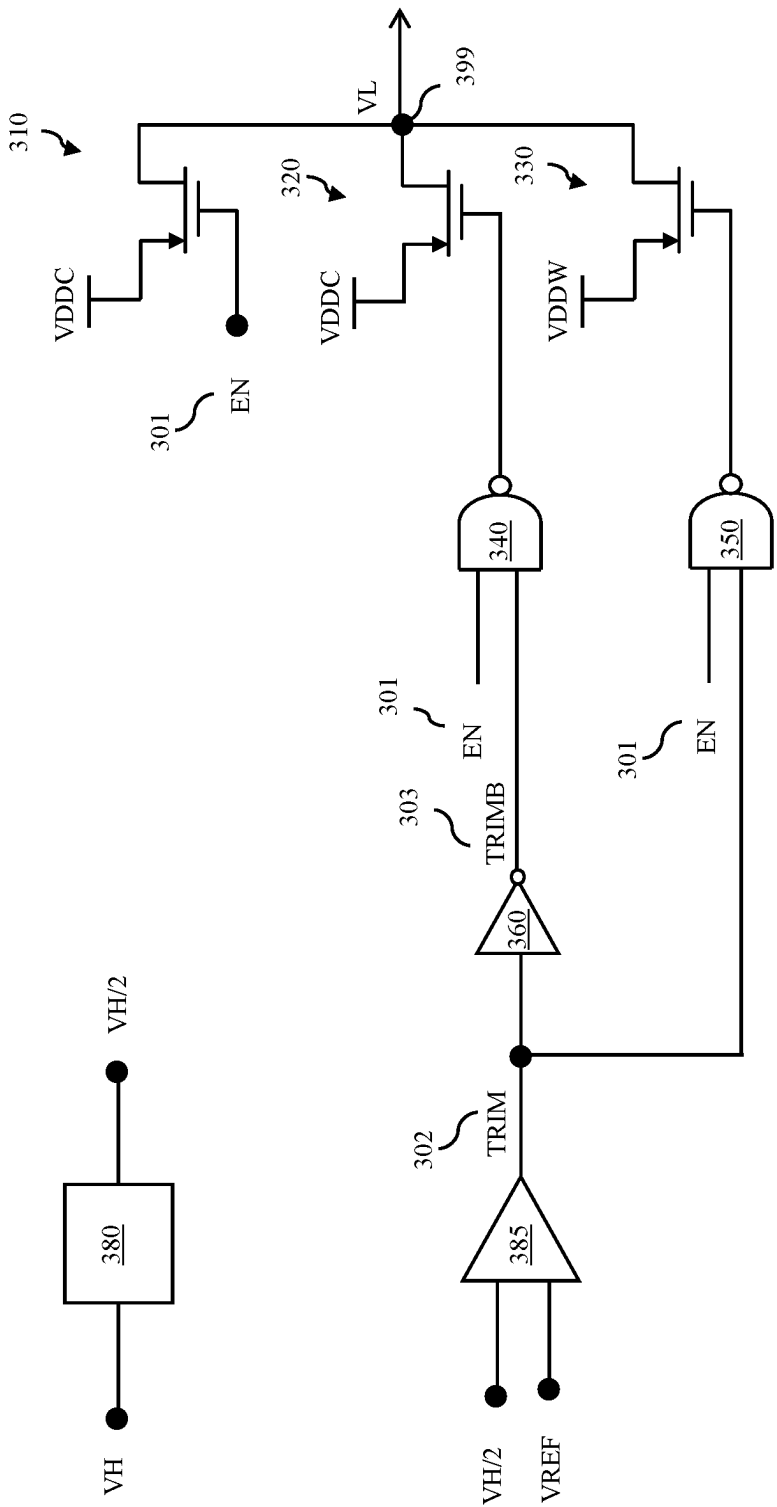


FIG. 3

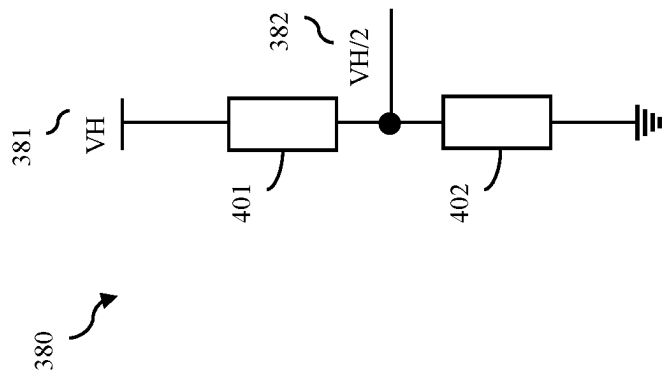


FIG. 4

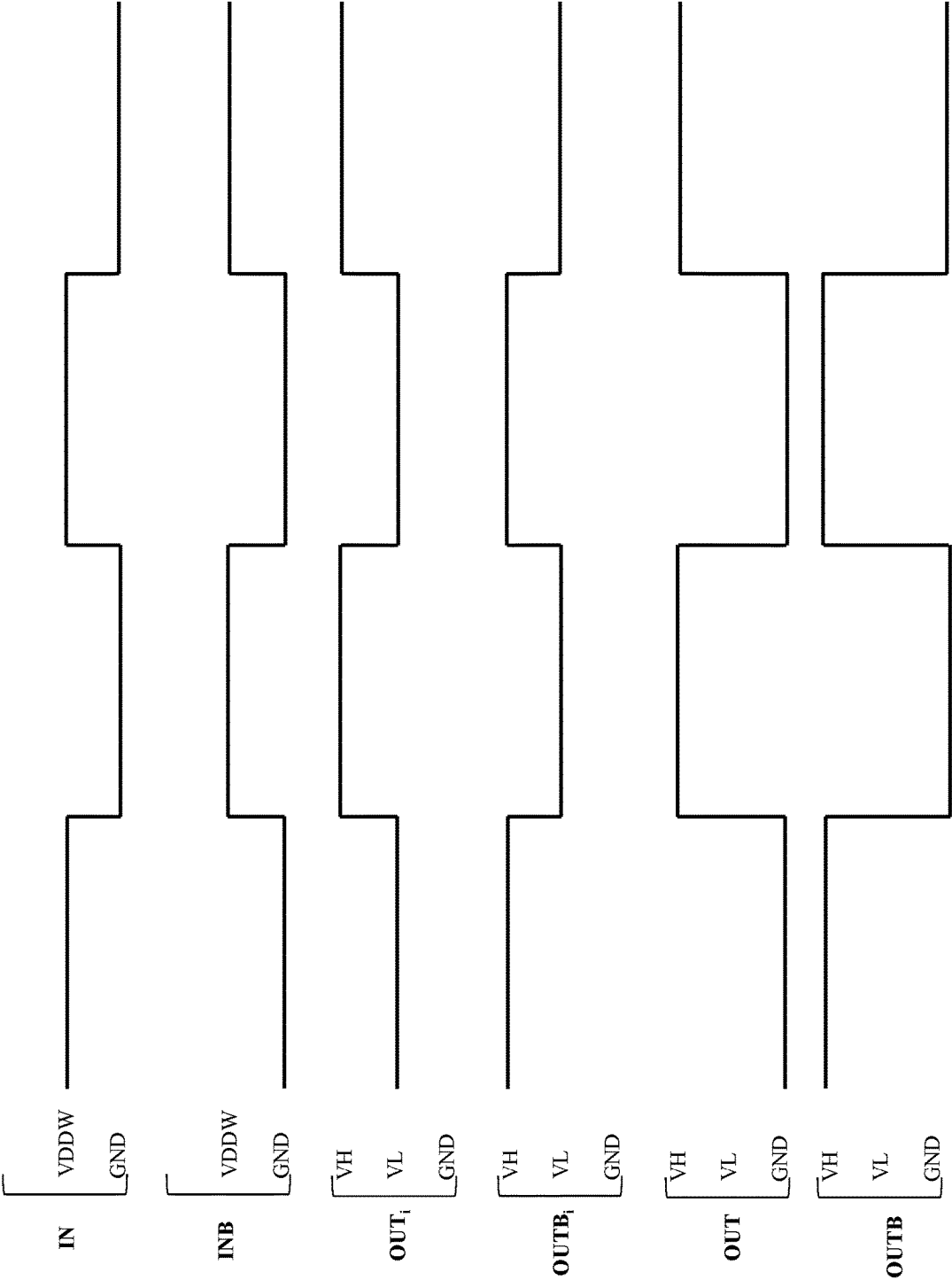


FIG. 5

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VOLTAGE LEVEL SHIFTER WITH PROGRAMMABLE HIGH SUPPLY VOLTAGE AND HIGH SUPPLY VOLTAGE-DEPENDENT VARIABLE LOW SUPPLY VOLTAGE

BACKGROUND

The present disclosure relates to on-chip voltage level shifters and, more particularly, to embodiments of a voltage level shifter with a programmable high positive supply voltage.

Considerations in modern integrated circuit (IC) design include, but are not limited to, performance improvement, size scaling, and power consumption. Oftentimes design changes to improve one aspect of the design can yield an undesirable trade-off with respect to other aspects of the design. For example, a voltage level shifter in an IC design may be configured to translate (i.e., shift) a voltage signal from one logic level or voltage domain to another. Voltage level shifter designs have been developed that include low voltage transistors to reduce power consumption. However, at least some of the low voltage transistors in these voltage level shifters tend to quickly operate outside the safe operating area (SOA). Design modifications intended to avoid operation outside the SOA have included, for example, the use of asymmetric high voltage transistors (e.g., laterally diffused metal oxide semiconductor field effect transistors (LDMOSFETs)) in place of at-risk low voltage transistors and/or the use of a discrete biasing circuit. While such design modifications may help to avoid operation outside the SOA, they also add to circuit complexity, power consumption, and area.

SUMMARY

Disclosed herein are embodiments of a structure. The structure includes a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure further includes a programmable voltage generator that outputs a high supply voltage to the high supply voltage node and a variable voltage generator that outputs a low supply voltage to the low supply voltage node. The low supply voltage output by the variable voltage generator can specifically be dependent on and lower than the high supply voltage.

In some embodiments, the structure can include a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure can further include a programmable voltage generator that receives a trim bit signal and, based on the trim bit signal, outputs a high supply voltage to the high supply voltage node. The structure can further include a variable voltage generator that outputs a low supply voltage to the low supply voltage node. In order to set the voltage level of the low supply voltage, the variable voltage generator can also receive the trim bit signal and, based on the trim bit signal, can generate and output the low supply voltage such that it is lower than the high supply voltage.

In other embodiments, the structure can include a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure can further include a programmable voltage generator that outputs a high supply voltage to the high supply voltage node. The structure can also include a variable voltage generator that outputs a low supply voltage to the low supply voltage node. In order to set the voltage level of the low supply voltage, the variable voltage generator can receive the high supply voltage,

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internally generate a trim bit signal based on the high supply voltage, and further generate and output the low supply voltage based on the trim bit signal so it is lower than the high supply voltage.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will be better understood from the following detailed description with reference to the drawings, which are not necessarily drawn to scale and in which: FIG. 1 is a circuit diagram illustrating embodiments of a disclosed structure;

FIG. 2 is a circuit diagram illustrating an example of a variable voltage generator that can be incorporated into the structure of FIG. 1;

FIG. 3 is a circuit diagram illustrating another example of a variable voltage generator that can be incorporated into the structure of FIG. 1;

FIG. 4 is a circuit diagram illustrating an example of a voltage divider that can be incorporated into the variable voltage generator of FIG. 3; and

FIG. 5 is a timing diagram illustrating input voltage pulses to and output voltage pulses from the structure of FIG. 1.

DETAILED DESCRIPTION

As mentioned above, considerations in modern integrated circuit (IC) design include, but are not limited to, performance improvement, size scaling, and power consumption. Oftentimes design changes to improve one aspect of the design can yield an undesirable trade-off with respect to other aspects of the design. For example, a voltage level shifter in an IC design may be configured to translate (i.e., shift) a voltage signal from one logic level or voltage domain to another. Voltage level shifter designs have been developed that include low voltage transistors to reduce power consumption. However, at least some of the low voltage transistors in these voltage level shifters tend to quickly operate outside the safe operating area (SOA). Design modifications intended to avoid operation outside the SOA have included, for example, the use of asymmetric high voltage transistors (e.g., laterally diffused metal oxide semiconductor field effect transistors (LDMOSFETs)) in place of at-risk low voltage transistors and/or the use of a discrete biasing circuit. While such design modifications may help to avoid operation outside the SOA, they also add to circuit complexity, power consumption, and area.

In view of the foregoing, disclosed herein are embodiments of a circuit structure that includes a voltage level shifter, a variable voltage generator, and a programmable voltage generator. The voltage level shifter can include a low supply voltage node, a high supply voltage node, and parallel branches, including multiple transistors, connected between the low and high supply voltage nodes. The variable voltage generator can generate a variable low supply voltage (VL) and can apply VL to the low supply voltage node. The programmable voltage generator can generate a programmable high supply voltage (VH) and can apply VH to the high supply voltage node. The VH can be any of multiple (e.g., up to three) different VH levels (e.g., a relatively low first VH level of 1.8 volts (V), a relatively high second VH level of 3.6V, and a mid-level third VH level between the first and second VH levels, such as at 2.6V). The variable voltage generator can be configured to track the voltage level of VH and to adjust the voltage level of VL based thereon. In some embodiments, such tracking can be performed using a trim bit signal employed by the

programmable voltage generator (e.g., a charge pump thereof) when programming VH. In other embodiments such tracking can be achieved by direct monitoring of VH. In either case, VL can be lower than VH and the variable voltage generator can increase VL from a relatively low first VL level (e.g., of 0.8V) to a relatively high second VL level (e.g., of 1.8 volts (V)) in response to a relatively high second VH (e.g., of 3.6V) to ensure operation within the safe operating area (SOA). The variable voltage generator can further decrease VL from the second VL level (e.g., of 1.8V) back to the first VL level (e.g., of 0.8V) when VH drops to the first VH level (e.g., of 1.8V) to ensure that the voltage level shifter remains operable and, optionally, even when VH drops to the mid-level third VH level (e.g., of 2.6V) to improve operating speed.

FIG. 1 is a schematic diagram generally illustrating embodiments of a circuit structure **199** including a voltage level shifter **100**, a programmable voltage generator **155**, and a variable voltage generator **150**.

The voltage level shifter **100** can, for example, include multiple transistors. As discussed in greater detail below, these transistors can include both P-type transistors and N-type transistors. For purposes of illustration, the P-type transistors and N-type transistors are described below and illustrated in the figures as being metal oxide semiconductor field effect transistors (MOSFETs) and, particularly, P-type field effect transistors (PFETs) and N-type field effect transistors (NFETs), respectively. A MOSFET refers to a transistor with a semiconductor channel region positioned laterally between a source region and a drain region and with a gate (e.g., including a gate dielectric-gate conductor stack) adjacent to the channel region. However, the figures and discussion thereof are not intended to be limiting. For example, alternatively, a similar circuit structure could be formed using bipolar junction transistors (BJTs) and, particularly, PNP BJTs and NPN BJTs. The transistors can, for example, all have the same maximum voltage rating (VDDW) and, for reduced power consumption, VDDW can be relatively low as specified by the transistor data sheet. For example, the transistors can have a maximum voltage rating of 3.3V or lower. In some embodiments, the transistors can have a maximum voltage rating of 1.8V, 1.5V or 0.8V. To minimize manufacturing complexity, the transistors can all be symmetric. That is, they can be designed so that the source region and the drain region are the same (e.g., same size, doping, etc.) except for minor process variations and so that the same maximum voltage rating applies to the gate-source voltage (VGS), the gate-drain voltage (VGD), and the drain-source voltage (VDS). As discussed in greater detail below, the voltage level shifter **100** can specifically be configured to achieve voltage level shifting without violating any of these maximum voltage ratings in order to avoid device stress and ensure operation within the SOA.

For purposes of illustration, examples provided herein are generally based on all transistors having a 1.8V maximum voltage rating. Additionally, various components of the circuit structure can, as discussed in greater detail below, be connected to receive fixed supply voltages. The fixed supply voltages available to the circuit structure can include but are not limited to: a fixed supply voltage (VDDW), which is set at a voltage level equal to the maximum voltage rating of the transistors (e.g., VDDW=1.8V), and an additional fixed supply voltage (VDDC), which is set at a voltage level between ground and VDDW. For example, VDDC can be essentially $\frac{1}{2}$ the magnitude of VDDW, less than $\frac{1}{2}$ the magnitude of VDDW, etc. For example, if VDDW=1.8V, VDDC=0.8V.

The programmable voltage generator **155** can be configured to generate and output a programmable high supply voltage (VH) and the variable voltage generator **150** can be configured to generate and output a variable VH-dependent low supply voltage (VL). The voltage level shifter **100** can include a high supply voltage node, which is electrically connected to the programmable voltage generator **155** for receiving VH, and a low supply voltage node **191**, which is electrically connected to the variable voltage generator **150** for receiving VL.

More specifically, the programmable voltage generator **155** can be configured to output VH at a relatively low first VH level, a relatively high second VH level, and, optionally, at some medium-level third VH level (between the first and second VH levels). The programmable voltage generator **155** can be configured so that the second VH level is essentially double the magnitude of VDDW, so that the first VH level is equal to VDDW and, if applicable, so that the third VH level is somewhere between the first and second VH levels. For example, in some embodiments, VDDW is 1.8V, the first VH level is 1.8V, the second VH level is 3.6V, and, if applicable, the third VH level is 2.6V. For purposes of this disclosure, “essentially double” the magnitude refers to twice the magnitude, plus or minus 10% thereof (e.g., if VDDW is 1.8V, then the high VH=3.6V $\pm$ 0.36V, etc.). Programmable voltage generators configured to output different supply voltages are well known in the art and, thus, the details thereof have been omitted from this specification in order to allow the reader to focus on the salient aspects of the disclosed embodiments. However, those skilled in the art will recognize that, typically, a programmable voltage generator will include a charge pump configured to convert an input direct current (DC) power supply to one or more different DC power supplies and the state of a trim bit signal (TRIM) received by the programmable voltage generator will dictate the level of VH output to a particular supply voltage node (in this case to the high supply voltage node **192**). For example, for a two-VH level programmable voltage generator, a single-bit two-state TRIM can control whether a high VH or a low VH is output to the high supply voltage node **192**. When TRIM is at a high logic level (i.e., a “1” logic level or a logic 1), the high VH can be output to the second supply node **192** and, when TRIM is at a low logic level (i.e., a “0” logic level or logic 0), the low VH can be output to the second voltage node **192**. For a three-VH programmable voltage generator, a tri-state single-bit TRIM can control whether a high VH, a mid-level VH, or a low VH is output to the high supply voltage node **192**. When TRIM is at logic 1, the high VH can be output to the second supply node **192**. When TRIM is at logic 0, the mid-level VH can be output to the second voltage node **192**. When TRIM is at a high impedance (i.e., high Z) state, the low VH can be output to the second voltage node **192**.

The variable voltage generator **150** can be configured to track the voltage level of VH and generate and output VL depending upon the voltage level of VH (i.e., to generate and output a VH-dependent VL) in order to, for example, ensure voltage level shifter transistor operation within the SOA when VH is high, to ensure voltage level shifter operability when VH is low and, if applicable, to improve operating speed when VH is at some mid-level. In some embodiments, such tracking can be performed by the variable voltage generator **150** using an external trim bit signal (TRIM) employed by the programmable voltage generator **155** when programming VH. In another embodiment such tracking can be achieved by direct monitoring of VH.

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FIG. 2 is a schematic diagram illustrating one example of a variable voltage generator **150A** that can be incorporated into circuit structure **199** of FIG. 1 for supplying a VH-dependent VL to the low supply voltage node **191** of the voltage level shifter. In this example, the variable voltage generator **150A** is configured to track VH using TRIM **202**, which is externally generated and also employed by the programmable voltage generator **155** to establish the voltage level of VH. The variable voltage generator **150A** is further configured to generate and output VL at a first VL level (e.g., at 0.8V) or at a higher second VL level (e.g., at 1.8V) depending upon the state of TRIM **202** as well as the state of an enable signal (EN) **201**.

In the disclosed embodiments, the circuit structure **199** can, for example, be incorporated into a memory circuit, such as a resistive random access memory (RRAM) circuit. In this case, EN can be a write enable signal, which is high during a write mode, and which is low during a read mode. For example, when EN **201** is at logic 0 and TRIM **202** is at a high-Z state (e.g., indicating that the programmable voltage generator **155** is outputting VH at the low first VH level, such as at 1.8V) or when EN **201** is at logic 1 and TRIM is at logic 0 (e.g., indicating that the programmable voltage generator **155** is outputting VH at a medium third VH level, such as a 2.6V), the variable voltage generator **150A** can output VL at the low first VL level (e.g., at 0.8V). However, when EN **201** and TRIM **202** are both at logic 1 (e.g., indicating that the programmable voltage generator **155** is outputting VH at the high second VH level, such as at 3.6V), the variable voltage generator **150A** can output VL at the high second VL level (e.g., at 1.8V).

More specifically, the variable voltage generator **150A** can include: an inverter **260**; a pair of NAND gates (i.e., a first NAND gate **240** and a second NAND gate **250**); three PFETs (i.e., a first PFET **210**, a second PFET **220**, and a third PFET **230**); and an output voltage node **299**. The output voltage node **299** can be electrically connected to the low supply voltage node **191** of the voltage level shifter **100** for supplying VL to the low supply voltage node **191**. The first PFET **210** can include: a source region, which is electrically connected to receive VDDC; a drain region, which is electrically connected to the output voltage node **299**; and a gate, which is electrically connected to receive EN **201**. The second PFET **220** can include: a source region, which is electrically connected to receive VDDC; a drain region, which is electrically connected to the output voltage node **299**; and a gate, which is electrically connected to receive an output signal from the first NAND gate **240**. The third PFET **230** can include: a source region, which is connected to receive VDDW; a drain region, which is connected to the output voltage node **299**; and a gate, which is electrically connected to receive an output signal from the second NAND gate **250**. The inverter **260** can be electrically connected to receive, as an input, the externally generated TRIM **202** and can be configured to output an inverted trim bit signal (TRIMB) **203** (i.e., TRIMB **203** can be inverted with respect to TRIM **202**). The first NAND gate **240** can be electrically connected to receive, as inputs, EN **201** and TRIMB **203**, respectively, and can be configured to output a first NAND gate output signal (according to a conventional NAND gate truth table). The second NAND gate **250** can be electrically connected to receive, as inputs, EN **201** and TRIM **202**, respectively, and can be configured to output a second NAND gate output signal (according to a conventional NAND gate truth table).

Those skilled in the art will recognize that a conventional NAND gate truth table specifies that when the inputs to a

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NAND gate are both low (i.e., 00) or are different (e.g., 01 or 10), then the NAND gate output signal will be high. When the inputs to the NAND gate are both at high (i.e., 11), then the NAND gate output signal will be low (i.e., 0).

In operation, the on/off state of the first PFET **210** is controlled by EN **201**, the on/off state of the second PFET **220** is controlled by the output signal from the first NAND gate **240**, and the on/off state of the third PFET **230** is controlled by the output signal from the second NAND gate **250**. When EN is low, the PFET **210** will turn on. Additionally, the output signals from the first and second NAND gates **240** and **250** will both be high so the second PFET **220** and the third PFET **230** will be turned off. Thus, a VL at the low first VL level of VDDC (e.g., at 0.8V) will be supplied by the first PFET **210** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. When EN **201** is high, TRIM **202** is low, and TRIMB **203** is high, the first PFET **210** will turn off, the output signal from the second NAND gate **250** will be high so the third PFET **230** will turn off, and the output signal from first NAND gate **240** will be low so the second PFET **220** will turn on. Thus, a VL at the low first VL level of VDDC is supplied through the second PFET **220** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. Finally, when EN **201** is high, TRIM **202** is high, and TRIMB **203** is low, the first PFET **210** is off, the output signal from the first NAND gate **240** will be high so the second PFET **220** is off, and the output signal from the second NAND gate **250** will be low so the third PFET **230** turns on. Thus, a VL at the high second VL level of VDDW (e.g., at 1.8V) is supplied through the third PFET **230** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**.

FIG. 3 is a schematic diagram illustrating another example of a variable voltage generator **150B** that can be incorporated into the circuit structure **199** of FIG. 1 for supplying a VH-dependent VL to the low supply voltage node **191** of the voltage level shifter **100**. In this example, the variable voltage generator **150B** is specifically configured to track VH through direct monitoring.

Specifically, the variable voltage generator **150B** can include: a voltage divider **380**; a comparator **385**; an inverter **360**; a pair of NAND gates (i.e., a first NAND gate **340** and a second NAND gate **350**); three PFETs (i.e., a first PFET **310**, a second PFET **320**, and a third PFET **330**); and an output voltage node **399**. The output voltage node **399** can be electrically connected to the low supply voltage node **191** of the voltage level shifter **100** for supplying the VL to the low supply voltage node **191**.

Voltage divider **380** can be connected to receive VH from the programmable voltage generator **155**. The voltage divider **380** can further be configured to divide VH essentially in half (e.g., half+/-10%) and, more particularly, to output an output voltage that is essentially half the magnitude of VH (i.e., VH/2). FIG. 4 is a schematic diagram illustrating an example of a voltage divider **380** that can be incorporated into the variable voltage generator **150B**. As illustrated, this simple voltage divider can include a pair of resistors **401**, **402** connected in series between an input node for receiving VH and ground. VH/2 can be output at the junction between the two resistors **401**, **402**. The voltage divider shown in FIG. 4 is provided for illustration purposes. Alternatively, any other suitable voltage divider circuit could be employed to generate and output VH/2 from VH.

Comparator **385** can be connected to receive, as inputs, VH/2 and a reference voltage (VREF), respectively. VREF

can be at some predetermined voltage level between the high second VH level divided by two and the low first VH level divided by two. For example, if the high second VH level is 3.6V so that high VH/2 is 1.8V and if the medium third VH level at 2.6V so that the medium third VH level divided by two is 1.3V, then VREF can be equal to or greater than 1.3V but less than 1.8V. Comparator **385** can further be configured to compare VH/2 to VREF and to internally generate and output a trim bit signal (TRIM **302**) based on the results of the comparison. Specifically, comparator **385** can output TRIM **302** at a logic 0, when VH/2 is less equal to or less than VREF, whereas comparator **385** can output TRIM **302** at a logic 1, when VH/2 is greater than VREF.

The remaining components of the variable voltage generator **150B** can be essentially the same as the components of the variable voltage generator **150A**, described above. Specifically, the first PFET **310** can include: a source region, which is electrically connected to receive VDDC (e.g., at 0.8V); a drain region, which is electrically connected to the output voltage node **399**; and a gate, which is electrically connected to receive EN **301**. The second PFET **320** can include: a source region, which is electrically connected to receive VDDC; a drain region, which is electrically connected to the output voltage node **399**; and a gate, which is electrically connected to receive an output signal from the first NAND gate **340**. The third PFET **330** can include: a source region, which is connected to receive VDDW; a drain region, which is connected to the output voltage node **399**; and a gate, which is electrically connected to receive an output signal from the second NAND gate **350**. The inverter **360** can be electrically connected to receive, as an input, TRIM **302** (which in this case has been internally generated) and can be configured to output an inverted trim bit signal (TRIMB) **303** (i.e., TRIMB **303** is inverted with respect to TRIM **302**). The first NAND gate **340** can be electrically connected to receive, as inputs, EN **301** and TRIMB **303**, respectively, and can be configured to output a first NAND gate output signal (according to a conventional NAND gate truth table). The second NAND gate **350** can be electrically connected to receive, as inputs, EN **301** and TRIM **302**, respectively, and can be configured to output a second NAND gate output signal (according to a conventional NAND gate truth table).

In operation, the on/off state of the first PFET **310** is controlled by EN **301**, the on/off state of the second PFET **320** is controlled by the output signal from the first NAND gate **340**, and the on/off state of the third PFET **330** is controlled by the output signal from the second NAND gate **350**. When EN **301** is low, the first PFET **310** turns on. Additionally, the output signals from the first and second NAND gates **340** and **350** will both be high so the second PFET **320** and the third PFET **330** will be off. Thus, a VL at the low first VL level of VDDC (e.g., at 0.8V) will be supplied by the first PFET **310** to the output voltage node **399** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. When EN **301** is high, TRIM **302** is low and TRIMB **303** is high, the first PFET **310** is off the output signal from the second NAND gate **350** is high so the third PFET **330** is off, and the output signal from first NAND gate **340** is low so the second PFET **320** turns on. Thus, a VL at the low first VL level of VDDC (e.g., at 0.8V) is supplied through the second PFET **320** to the output voltage node **399** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. Finally, when EN **301** is high, TRIM **302** is high, and TRIMB **303** is low, the first PFET **310** turns off, the output signal from the first NAND gate **340** is high so the second PFET **320** turns off, and the output

signal from the second NAND gate **350** is low so the PFET **330** turns on. Thus, a VL at the high second VL level of VDDW (e.g., at 1.8V) is supplied through the third PFET **330** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**.

Referring again to FIG. 1, the voltage level shifter **100** can include symmetric branches **101a-101b** connected in parallel between the low supply voltage node **191** and the high supply voltage node **192**.

Each branch **101a, 101b** can include a first PFET **110a, 110b** and a second PFET **120a, 120b** connected in series between low supply voltage node **191** and high supply voltage node **192**. That is, first PFET **110a, 110b** can include a drain region **113a, 113b** electrically connected to low supply voltage node **191**, a gate **112a, 112b**, and a source region **111a, 111b**. Second PFET **120a, 120b** can include a drain region **123a, 123b** electrically connected to source region **111a, 111b** of first PFET **110a, 110b**, a gate **122a, 122b**, and a source region **121a, 121b** electrically connected to high supply voltage node **192**. In some embodiments, source region **111a, 111b** of first PFET **110a, 110b** and drain region **123a, 123b** of the second PFET **120a, 120b** can be a shared source/drain region. In other embodiments, source region **111a, 111b** of the first PFET **110a, 110b** and drain region **123a, 123b** of the second PFET **120a, 120b** could be discrete regions electrically connected by an interconnect.

Each branch **101a, 101b** can further include an intermediate output node **125a, 125b** and a third PFET **130a, 130b** electrically connected to a junction **116a, 116b** between first PFET **110a, 110b** and second PFET **120a, 120b**. Third PFET **130a, 130b** can specifically include: a source region **131a, 131b**, which is electrically connected to junction **116a, 116b** between first PFET **110a, 110b** and second PFET **120a, 120b**; a drain region **133a, 133b**, which is electrically connected to a final output node **135a, 135b** and also to gate **112a, 112b** of first PFET **110a, 110b**, and a gate **132a, 132b**, which is electrically connected to, e.g., low supply voltage node **191**, so as to receive VL.

Additionally, the second PFET **120a** of branch **101a** and second PFET **120b** of branch **101b** can be cross-coupled. That is, gate **122a** of second PFET **120a** in branch **101a** can be electrically connected to junction **116b** between first PFET **110b** and second PFET **120b** in branch **101b**, while gate **122b** of second PFET **120b** in branch **101b** can be electrically connected to junction **116a** between first PFET **110a** and second PFET **120a** in branch **101a**.

Finally, each branch **101a, 101b** can further include an NFET **140a, 140b** (also referred to herein as a pass-gate NFET). NFET **140a, 140b** can include: a source region **141a, 141b**, which is electrically connected to an input node **145a, 145b**; a drain region **143a, 143b**, which is electrically connected to final output node **135a, 135b** (and thereby electrically connected to the drain region **133a, 133b** of third PFET **130a, 130b** as well as gate **112a, 112b** of first PFET **110a, 110b**); and a gate, which is electrically connected to receive a pass-gate control voltage that is, for example, equal to the maximum voltage rating (VDDW) of the transistors therein (e.g., 1.8V).

The voltage level shifter **100** can further be connected to receive two input voltage pulses at the two input nodes **145a** and **145b**, respectively. Input voltage pulse (IN) at input node **145a** can transition between VDDW (e.g., at 1.8V) and ground (GND) (e.g., at 0.0V). An inverted input voltage pulse (INB) at input node **145b** can be inverted with respect to IN and, thus, can transition from GND to VDDW as IN transitions from VDDW to GND and vice versa.

For purposes of this disclosure, a “voltage pulse” refers to an essentially rectangular direct current voltage signal that transitions between a low voltage level and a high voltage level at regular and repeated intervals such that the pulse is at the low voltage level for a first time period (T1), switches to the high voltage level and remains at the high voltage level for a second time period (T2) (which is the same as or different from T1), switches back to the low voltage level and again remains at the low voltage level for T1, and so on.

The voltage level shifter 100 can further be configured to generate and output four different output voltage pulses including two intermediate output voltage pulses at intermediate output nodes 125a-125b and two final output voltage pulses at final output nodes 135a-135b. The two intermediate output voltage pulses can include intermediate output voltage pulse (OUT_i), which will transition between VL and VH, at intermediate output node 125b and inverted intermediate output voltage pulse (OUTB_i), which is inverted with respect to OUT_i and, thus, will transition from VH to VL as OUT_i transitions from VL to VH and vice versa, at intermediate output node 125a. The two final output voltage pulses can include final output voltage pulse (OUT), which transitions between ground and VH, at a final output node 135b and inverted final output voltage pulse (OUTB), which is inverted with respect to OUT and, thus, which transitions from VH to ground as OUT transitions from ground to VH and vice versa, at final output node 135a.

FIG. 5 shows an example of a timing diagram illustrating input voltage pulses (IN, INB) to a voltage level shifter 100 and output voltage pulses (OUT_i, OUTB_i, OUT, OUTB) from the voltage level shifter 100, when the voltage level of VH is programmable (e.g., to 3.6V, 2.6V, or 1.8V) and when the voltage level of VL is automatically adjusted based on VH (e.g., when VH=3.6V, VL=1.8V; when VH=2.6V, VL=0.8V; and when VH=1.8V, VL=0.8V).

In operation, VDDW is continuously applied to gates 142a, 142b of NFETs 140a, 140b, VL is continuously applied to gates 132a, 132b of third PFETs 130a, 130b. When IN transitions from GND to VDDW and INB transitions from VDDW to GND, in branch 101b, gate and source voltages of NFET 140b are at VDDW and GND, respectively, so the threshold voltage (VT) of NFET 140b is reached. NFET 140b turns on and pulls down the voltage on final output node 135b to GND so OUT is at GND. When the voltage on final output node 135b is at GND, gate voltage on first PFET 110b is at GND and the VT of first PFET 110b is reached. First PFET 110b turns on, pulling down the voltage at intermediate output node 125b to VL so OUT_i is at VL. The gate and source voltages on third PFET 130b are both at VL, so third PFET 130b remains off. In branch 101a the gate voltage on second PFET 120a is at VL and the source voltage is at VH, so the VT of second PFET 120a is reached. Second PFET 120a turns on, thereby pulling up the voltage on intermediate output node 125a to VH so OUTB_i goes to VH. When intermediate output node 125a is at VH, the second PFET 120b remains off. Furthermore, the gate and source voltages on third PFET 130a are VL and VH, respectively, so the VT of third PFET 130a is reached. Third PFET 130a turns on, pulling up the voltage on final output node 135a to VH so OUTB goes to VH. Third PFET 130a also pulls the gate voltage of first PFET 110a up to VH, so first PFET 110a remains off. Since the gate and source voltages of NFET 140a are both at VDDW, NFET 140a also remains off.

When IN transitions from VDDW to GND and INB transitions from GND to VDDW, the reverse occurs. In branch 101a, gate and source voltages on NFET 140a are at

VDDW and GND, respectively, so the VT of NFET 140a is reached. NFET 140a turns on and pulls down the voltage on final output node 135a to GND so OUTB is at GND. When the voltage on final output node 135a is GND, the gate voltage on first PFET 110a is at GND and the VT of first PFET 110a is reached. First PFET 110a turns on, pulling down the voltage at intermediate output node 125a to VL so OUTB_i is at VL. The gate and source voltages of third PFET 130a are both at VL, so third PFET 130a remains off. Additionally, in branch 101b the gate voltage on second PFET 120b is at VL and the source voltage is at VH, so the VT of second PFET 120b is reached. Second PFET 120b turns on, thereby pulling up the voltage on intermediate output node 125b to VH so OUT_i goes to VH. When intermediate output node 125b is at VH, second PFET 120a remains off. Furthermore, the gate and source voltages on third PFET 130b are VL and VH, respectively, so the VT of third PFET 130b is reached. Third PFET 130b turns on, pulling up the voltage on final output node 135b to VH so OUT goes to VH. Third PFET 130b also pulls up the gate voltage on first PFET 110b to VH, so first PFET 110b remains off. Since the gate and source voltages on NFET 140b are both at VDDW, NFET 140b also remains off.

Thus, as illustrated, OUT_i at intermediate output node 125b transitions between VL and VH and back (regardless of whether VL is at 1.8V and VH is at 3.6V or VL is at 0.8V and VH is at either 2.6V or 1.8V) and OUTB_i at intermediate output node 125a is inverted with respect to OUT_i (i.e., transitions from VH to VL when OUT_i transitions from VL to VH and vice versa). Additionally, OUT at final output node 135b transitions between GND and VH and back (again regardless of whether VL is at 1.8V and VH is at 3.6V or VL is at 0.8V and VH is at either 2.6V or 1.8V) and OUTB at final output node 135a is inverted with respect to OUT (i.e., transitions from VH to GND when OUT transitions from GND to VH and vice versa).

In the disclosed circuit structure 199, the variable voltage generator 150A or 150B allows the voltage level of VL to be shifted upwards (e.g., from 0.8V to 1.8V) when the voltage level of VH is relatively high, e.g., at 3.6V. Thus, all transistors 110a-110b, 120a-120b, 130a-130b, 140a-140b that are incorporated into the voltage level shifter 100 can be low voltage (e.g., 1.8V) symmetric transistors without risking operation outside the SOA for such transistors. In the example provided, the output voltage pulses would be as follows: OUT_i transitions between 1.8V and 3.6V, OUTB_i is inverted with respect to OUT_i; OUT transitions between GND and 3.6V, and OUTB is inverted with respect to OUT. The variable voltage generator 150A or 150B also provides a means by which the voltage level of VL can be shifted downwards (e.g., from 1.8V to 0.8V) when, for whatever reason, a lower VH is desirable (e.g., to reduce power consumption). Those skilled in the art will recognize that in the above-described circuit structure, VH at a medium third VH level (e.g., of 2.6V) in combination with VL at a high second VL level (e.g., of 1.8V) would still leave the voltage level shifter operable; however, the switching speed of the transistors therein would be reduced. By automatically adjusting the voltage level of VL downwards (e.g., from 1.8V to 0.8V) when the medium third VH level (e.g., of 2.6V) is employed, the switching speed of the transistors is boosted, and performance is thereby improved. Those skilled in the art will also recognize that, in the above-described circuit structure, VH at a low first VH level (e.g., of 1.8V) in combination with VL at the high second VL level (e.g., of 1.8V) would leave the voltage level shifter inoperable because the threshold voltages of the transistors therein

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could not be reached. By automatically adjusting the voltage level of VL downward (e.g., from 1.8V to 0.8V) when VH is at the low first VH level (e.g., of 1.8 volts), the voltage level shifter remains operable.

In the above-described embodiments, body biasing of the transistors within the voltage level shifter **100** can also be employed, as indicated to FIG. 1, to improve transistor switching speeds.

It should be understood that the terminology used herein is for the purpose of describing the disclosed structures and methods and is not intended to be limiting. For example, as used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, as used herein, the terms “comprises,” “comprising,” “includes” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Furthermore, as used herein, terms such as “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “under,” “below,” “underlying,” “over,” “overlying,” “parallel,” “perpendicular,” etc., are intended to describe relative locations as they are oriented and illustrated in the drawings (unless otherwise indicated) and terms such as “touching,” “in direct contact,” “abutting,” “directly adjacent to,” “immediately adjacent to,” etc., are intended to indicate that at least one element physically contacts another element (without other elements separating the described elements). The term “laterally” is used herein to describe the relative locations of elements and, more particularly, to indicate that an element is positioned to the side of another element as opposed to above or below the other element, as those elements are oriented and illustrated in the drawings. For example, an element that is positioned laterally adjacent to another element will be beside the other element, an element that is positioned laterally immediately adjacent to another element will be directly beside the other element, and an element that laterally surrounds another element will be adjacent to and border the outer sidewalls of the other element. The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed.

The descriptions of the various disclosed embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limiting. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosed embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A structure comprising:
 - a voltage level shifter including a low supply voltage node and a high supply voltage node;
 - a programmable voltage generator outputting a high supply voltage to the high supply voltage node; and
 - a variable voltage generator outputting a low supply voltage to the low supply voltage node, wherein the low supply voltage is dependent on and lower than the high supply voltage.

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2. The structure of claim 1, wherein the voltage level shifter receives an input voltage pulse that transitions between an input voltage and ground and outputs multiple output voltage pulses including an intermediate output voltage pulse that transitions between the low supply voltage and the high supply voltage and a final output voltage pulse that transitions between ground and the high supply voltage.

3. The structure of claim 2,

wherein the voltage level shifter further includes multiple transistors, and

wherein the multiple transistors have the same voltage rating equal to the input voltage.

4. The structure of claim 3, wherein the multiple transistors are symmetric transistors.

5. The structure of claim 2,

wherein the programmable voltage generator is programmable to generate and output any of a first high supply voltage equal to the input voltage and a second high supply voltage essentially double the input voltage,

wherein, when the programmable voltage generator outputs the first high supply voltage, the variable voltage generator outputs a first low supply voltage lower than the input voltage, and

wherein, when the programmable voltage generator outputs the second high supply voltage, the variable voltage generator outputs a second low supply voltage equal to the input voltage.

6. The structure of claim 5,

wherein the programmable voltage generator is further programmable to output a third high supply voltage between the first high supply voltage and the second high supply voltage, and

wherein, when the programmable voltage generator outputs the third high supply voltage, the variable voltage generator outputs the first low supply voltage.

7. The structure of claim 2,

wherein the input voltage is 1.8 volts,

wherein the high supply voltage is programmable to any of a 1.8-volt high supply voltage and a 3.6-volt high supply voltage,

wherein, when the programmable voltage generator outputs the 1.8-volt high supply voltage, the variable voltage generator outputs a 0.8-volt low supply voltage, and

wherein, when the programmable voltage generator outputs the 3.6-volt high supply voltage, the variable voltage generator outputs a 1.8-volt low supply voltage.

8. The structure of claim 7,

wherein the high supply voltage is further programmable to a 2.6-volt high supply voltage, and

wherein, when the programmable voltage generator outputs the 2.6-volt high supply voltage, the variable voltage generator outputs the 0.8-volt low supply voltage.

9. The structure of claim 2, wherein the voltage level shifter includes:

two input nodes receiving two input voltage pulses, respectively, wherein the two input voltage pulses include the input voltage pulse and an inverted input voltage pulse; and

four output nodes outputting, in response to the two input voltage pulses, four output voltage pulses, respectively, wherein the four output voltage pulses include the final output voltage pulse, an inverted final output voltage pulse, the intermediate output voltage pulse, and an inverted intermediate output voltage pulse.

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10. The structure of claim 9,
 wherein the voltage level shifter further includes two
 branches connected between the low supply voltage
 node and the high supply voltage node, and
 wherein each branch includes:
 an input node;
 an intermediate output node;
 an output node;
 a first P-type transistor;
 a second P-type transistor, wherein the first P-type
 transistor and the second P-type transistor are con-
 nected in series between the low supply voltage node
 and the high supply voltage node and wherein the
 intermediate output node is at a junction between the
 first P-type transistor and the second P-type transis-
 tor;
 a third P-type transistor having a source region con-
 nected to the junction, a gate connected to receive the
 low supply voltage, and a drain region connected to
 a gate of the first P-type transistor and to the output
 node; and
 an N-type transistor having a source region connected
 to the input node, a gate connected to receive the
 input voltage, and a drain connected to the output
 node,
 wherein, between the two branches, second P-type
 transistors are cross-coupled.

11. A structure comprising:
 a voltage level shifter including a low supply voltage node
 and a high supply voltage node;
 a programmable voltage generator receiving a trim bit
 signal and, based on the trim bit signal, outputting a
 high supply voltage to the high supply voltage node;
 and
 a variable voltage generator outputting a low supply
 voltage to the low supply voltage node,
 wherein the variable voltage generator further receives the
 trim bit signal and generates and outputs the low supply
 voltage based on the trim bit signal, and
 wherein the low supply voltage is lower than the high
 supply voltage.

12. The structure of claim 11,
 wherein the voltage level shifter receives multiple input
 voltage pulses and, in response to the multiple input
 voltage pulses, outputs multiple output voltage pulses
 including an intermediate output voltage pulse that
 transitions between the low supply voltage and the high
 supply voltage and a final output voltage pulse that
 transitions between ground and the high supply volt-
 age,
 wherein the programmable voltage generator is program-
 mable to generate and output any of a first high supply
 voltage equal to an input voltage, a second high supply
 voltage essentially double the input voltage, and a third
 high supply voltage between the first high supply
 voltage and the second high supply voltage,
 wherein, when the programmable voltage generator out-
 puts either the first high supply voltage or the third high
 supply voltage, the variable voltage generator outputs a
 first low supply voltage lower than the input voltage,
 and
 wherein, when the programmable voltage generator out-
 puts the second high supply voltage, the variable volt-
 age generator outputs a second low supply voltage
 equal to the input voltage.

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13. The structure of claim 12, wherein the variable voltage
 generator includes:
 an output voltage node connected to the low supply
 voltage node;
 a first P-type transistor with a source region connected to
 receive the first low supply voltage, a drain region
 connected to the output voltage node, and a gate
 connected to receive an enable signal;
 a second P-type transistor with a source region connected
 to receive the first low supply voltage, a drain region
 connected to the output voltage node, and a gate
 connected to a first output of a first NAND gate;
 a third P-type transistor with a source region connected to
 receive the second low supply voltage, a drain region
 connected to the output voltage node, and a gate
 connected to a second output of a second NAND gate;
 and
 an inverter connected to receive the trim bit signal and
 configured to output an inverted trim bit signal,
 wherein the first NAND gate is connected to receive the
 enable signal and the inverted trim bit signal, and
 wherein the second NAND gate is connected to the enable
 signal and the trim bit signal.

14. The structure of claim 13,
 wherein the enable signal controls the first P-type tran-
 sistor,
 wherein output signals from the first NAND gate and the
 second NAND gate control the second P-type transistor
 and the third P-type transistor, respectively,
 wherein, when the enable signal is at a low logic level, the
 first P-type transistor turns on and the second P-type
 transistor and the third P-type transistor turn off, so that
 the first low supply voltage is supplied to the low
 supply voltage node of the voltage level shifter through
 the output voltage node,
 wherein, when the enable signal has a high logic level and
 the trim bit signal has the low logic level, the second
 P-type transistor turns on and the first P-type transistor
 and the third P-type transistor turn off, so that the first
 low supply voltage is supplied to the low supply
 voltage node of the voltage level shifter through the
 output voltage node, and
 wherein, when the enable signal and the trim bit signal
 have the high logic level, the first P-type transistor and
 the second P-type transistor turn off and the third P-type
 transistor turns on, so that the second low supply
 voltage is supplied to the low supply voltage node
 through the output voltage node.

15. The structure of claim 12,
 wherein the input voltage is 1.8 volts,
 wherein the high supply voltage is programmable to any
 of a 1.8-volt high supply voltage, a 2.6-volt high supply
 voltage, and a 3.6-volt high supply voltage,
 wherein, when the programmable voltage generator out-
 puts either the 1.8-volt high supply voltage or the
 2.6-volt high supply voltage, the variable voltage gen-
 erator outputs a 0.8-volt low supply voltage, and
 wherein, when the programmable voltage generator out-
 puts the 3.6-volt high supply voltage, the variable
 voltage generator outputs a 1.8-volt low supply volt-
 age.

16. A structure comprising:
 a voltage level shifter including a low supply voltage node
 and a high supply voltage node;
 a programmable voltage generator outputting a high sup-
 ply voltage to the high supply voltage node; and
 a variable voltage generator outputting a low supply
 voltage to the low supply voltage node,

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wherein the variable voltage generator receives the high supply voltage, internally generates a trim bit signal based on the high supply voltage, and generates and outputs the low supply voltage based on the trim bit signal, and

wherein the low supply voltage is lower than the high supply voltage.

17. The structure of claim 16,

wherein the voltage level shifter receives an input voltage pulse that transitions between an input voltage and ground and further outputs, in response to the input voltage pulse, multiple output voltage pulses including an intermediate output voltage pulse that transitions between the low supply voltage and the high supply voltage and a final output voltage pulse that transitions between ground and the high supply voltage,

wherein the high supply voltage is programmable to any of a first high supply voltage equal to the input voltage, a second high supply voltage essentially double the input voltage, and a third high supply voltage between the first high supply voltage and the second high supply voltage,

wherein, when the programmable voltage generator outputs either the first high supply voltage or the third high supply voltage, the variable voltage generator outputs a first low supply voltage lower than the input voltage, and

wherein, when the programmable voltage generator outputs the second high supply voltage, the variable voltage generator outputs a second low supply voltage equal to the input voltage.

18. The structure of claim 17, wherein the variable voltage generator includes:

a voltage divider receiving the high supply voltage and outputting a divided high supply voltage;

a comparator receiving the divided high supply voltage and a reference voltage and outputting the trim bit signal, wherein the trim bit signal is at a high logic level when the divided high supply voltage is greater than the reference voltage and at a low logic level when the divided high supply voltage is less than or equal to the reference voltage;

an output voltage node connected to the low supply voltage node;

a first P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to receive an enable signal;

a second P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to a first output of a first NAND gate;

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a third P-type transistor with a source region connected to receive the second low supply voltage, a drain region connected to the output voltage node, and a gate connected to a second output of a second NAND gate; and

an inverter receiving the trim bit signal and outputting an inverted trim bit signal,

wherein the first NAND gate receives the enable signal and the inverted trim bit signal, and

wherein the second NAND gate receives the enable signal and the trim bit signal.

19. The structure of claim 18,

wherein the enable signal controls the first P-type transistor,

wherein output signals from the first NAND gate and the second NAND gate control the second P-type transistor and the third P-type transistor, respectively,

wherein, when the enable signal has a low logic level, the first P-type transistor turns on and the second P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node,

wherein, when the enable signal has a high logic level and the trim bit signal has the low logic level, the second P-type transistor turns on and the first P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node, and

wherein, when the enable signal and the trim bit signal have the high logic level, the first P-type transistor and the second P-type transistor turn off and the third P-type transistor turns on, so that the second low supply voltage is supplied to the low supply voltage node through the output voltage node.

20. The structure of claim 17,

wherein the input voltage is 1.8 volts,

wherein the high supply voltage is programmable to any of a 1.8-volt high supply voltage, a 2.6-volt high supply voltage, and a 3.6-volt high supply voltage,

wherein, when the programmable voltage generator outputs either the 1.8-volt high supply voltage or the 2.6-volt high supply voltage, the variable voltage generator outputs a 0.8-volt low supply voltage, and

wherein, when the programmable voltage generator outputs the 3.6-volt high supply voltage, the variable voltage generator outputs a 1.8-volt low supply voltage.

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